

per unit of risk (standard deviation). I will also use the *raw Sharpe ratio*, sometimes called the reward-to-risk ratio, which refers to

$$\text{Raw Sharpe Ratio} = \frac{E(r)}{\sigma}, \quad (2.9)$$

which does not subtract the risk-free rate. In Figure 2.9, the raw Sharpe ratio for equities was 0.58.

The problem of optimal asset allocation over one risky asset and one risk-free asset is solved by finding the tangency point of the highest indifference curve with the CAL. Figure 2.9 shows two tangency points for risk aversion levels of $\gamma = 7$ (Panel A) and $\gamma = 2$ (Panel B). In the $\gamma = 7$ case, the investor holds 40% equity and 60% T-bills. The tangency point on the CAL is to the left of the equity position marked with the square. In the $\gamma = 2$ case, the asset owner holds a levered portfolio. This more risk-tolerant investor holds a much more aggressive equity position. He shorts the risk-free asset or borrows money. He holds -39% in risk-free assets and a 139% position in equities. The tangency point of the indifference curve is to the right-hand side of the equities position marked with the square.

The analytical solution in this simple setting of one risky asset and one risk-free asset for the optimal weight in the risky asset, w^* , is

$$w^* = \frac{1}{\gamma} \frac{E(r) - r_f}{\sigma^2}, \quad (2.10)$$

and $1 - w^*$ held in the risk-free asset. In equation (2.10), note that as the investor becomes more risk averse ($\gamma \rightarrow \infty$), the risky asset weight goes to zero and the investor holds entirely T-bills. As the risky asset becomes more attractive ($E(r) - r_f$ increases or σ decreases) the optimal weight in the risky asset also increases.

What is the market's risk aversion coefficient? At the end of December 2011, the combined stock market capitalizations on the NYSE and NASDAQ were \$11.8 and \$3.8 trillion, respectively—a total of \$15.6 trillion in equities. At the same date there were \$1.5 trillion in T-bills issued by the U.S. Treasury. Taking only T-bills as the risk-free asset, we have a weight of

$$w^* = \frac{15.6}{15.6 + 1.5} = 91.2\%$$

invested in equities by a “representative” investor who is assumed to represent the U.S. equity and T-bill markets. (I ignore all the non-U.S. risky markets and the risky fixed income markets for the purposes of this question.) Rearranging the optimal weight in equation (2.10) at a risk-free rate of 1% and using the equity premium and equity volatility over January 1926 to December 2011, we have

$$\gamma = \frac{1}{w^*} \frac{E(r) - r_f}{\sigma^2} = \frac{1}{0.9123} \frac{0.1019}{(0.1915)^2} = 3.0,$$